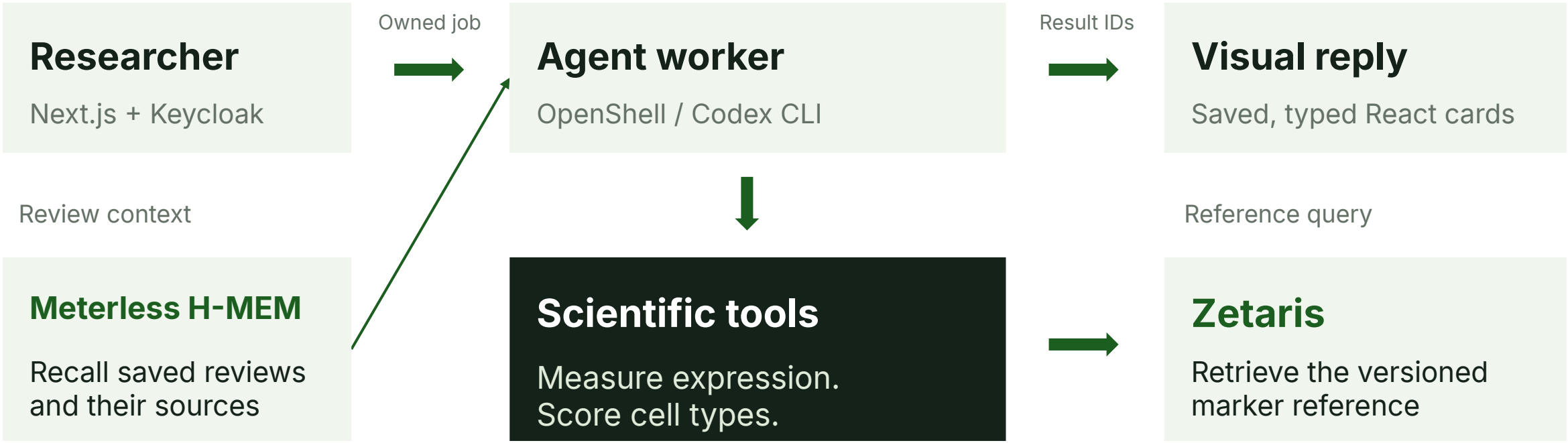
An agent connects the tools to the conversation.



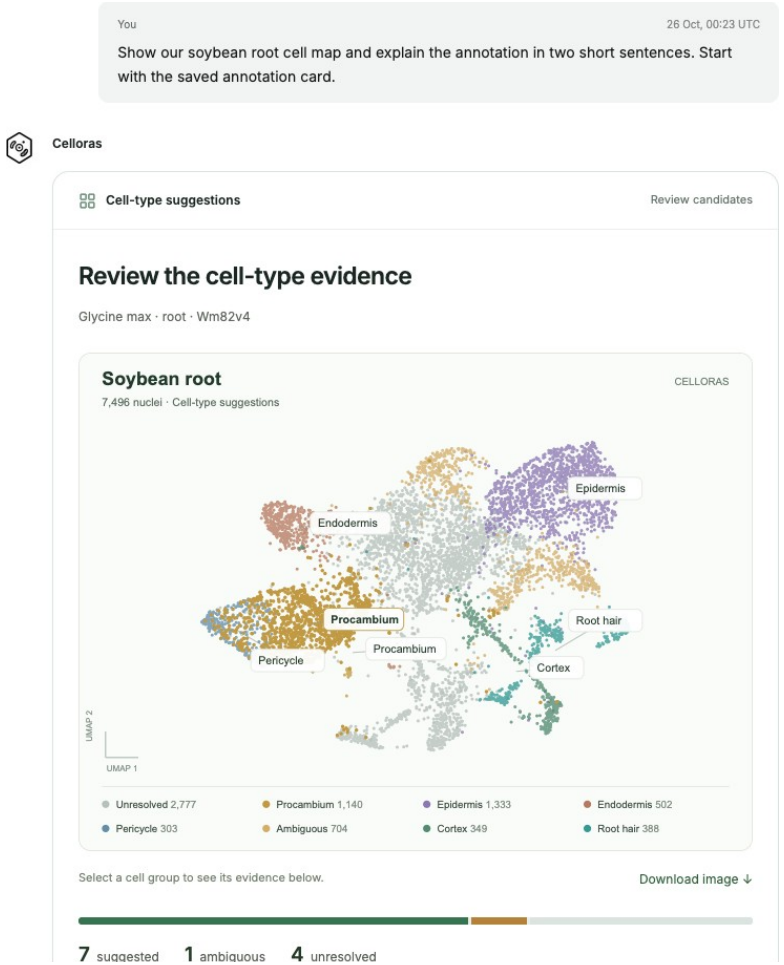
PostgreSQL: owned experiments, queued turns, results, reviews and memory history

A visual answer, with evidence to inspect.

The agent proposes cell types and explains why.

Its reply can contain the same interactive map and evidence cards used across Celloras.

You can challenge the result, review its sources, and save a correction in the conversation.



Each source answers a different part of the question.

MEASURED CELLS

GSE270392 root atlas

7,496 nuclei, RNA counts, original UMAP

MARKER REFERENCE

Published Table S2 via Zetaris

42 associations, 41 genes, 9 root cell types

RESEARCH CONTEXT

Curated papers and guide

3 sources, 68 searchable passages

Celloras agent

Compare measured activity with compatible marker evidence. Explain candidates and uncertainty.

Saved reviews enter separately through Meterless H-MEM.

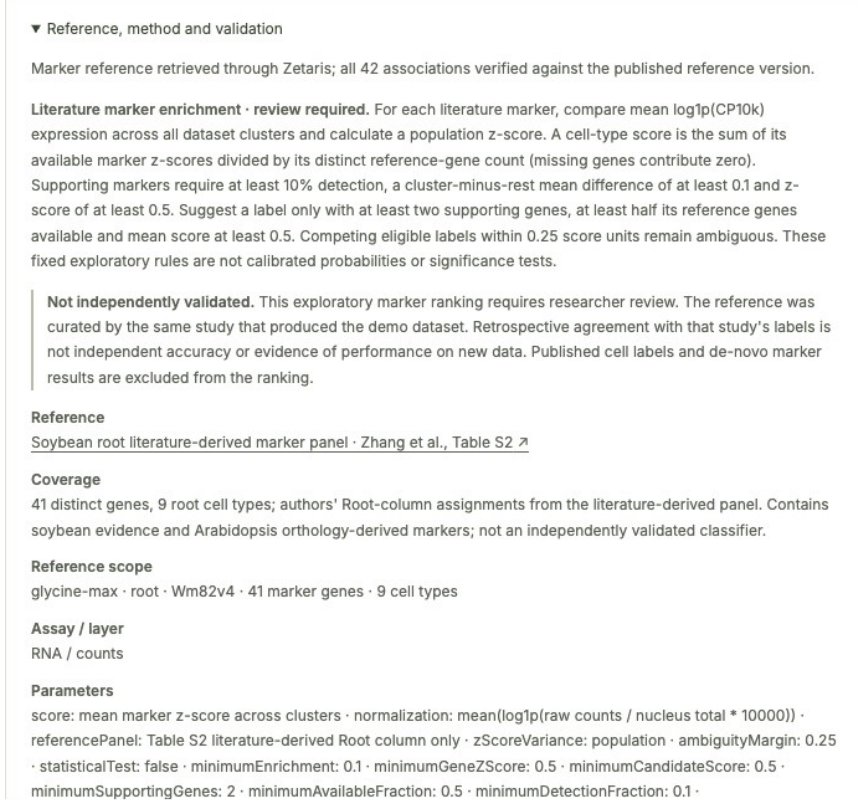
Next connections: H5AD uploads, SoyBase mapping, live orthology and nodule reference panels.

Zetaris supplies the reference used for annotation.

A real, read-only SQL connection

```
SELECT record_index, record_json, content_sha256
FROM CELLORAS.marker_reference_v1
WHERE reference_id = <pinned reference>
ORDER BY record_index
```

1 Retrieve all 42 associations.
2 Verify their order, count and content hash.
3 Save a query receipt with the result.



▼ Reference, method and validation

Marker reference retrieved through Zetaris; all 42 associations verified against the published reference version.

Literature marker enrichment · review required. For each literature marker, compare mean log1p(CP10k) expression across all dataset clusters and calculate a population z-score. A cell-type score is the sum of its available marker z-scores divided by its distinct reference-gene count (missing genes contribute zero). Supporting markers require at least 10% detection, a cluster-minus-rest mean difference of at least 0.1 and z-score of at least 0.5. Suggest a label only with at least two supporting genes, at least half its reference genes available and mean score at least 0.5. Competing eligible labels within 0.25 score units remain ambiguous. These fixed exploratory rules are not calibrated probabilities or significance tests.

> **Not independently validated.** This exploratory marker ranking requires researcher review. The reference was curated by the same study that produced the demo dataset. Retrospective agreement with that study's labels is not independent accuracy or evidence of performance on new data. Published cell labels and de-novo marker results are excluded from the ranking.

Reference
Soybean root literature-derived marker panel · Zhang et al., Table S2 ↗

Coverage
41 distinct genes, 9 root cell types; authors' Root-column assignments from the literature-derived panel. Contains soybean evidence and Arabidopsis orthology-derived markers; not an independently validated classifier.

Reference scope
glycine-max · root · Wm82v4 · 41 marker genes · 9 cell types

Assay / layer
RNA / counts

Parameters
score: mean marker z-score across clusters · normalization: mean(log1p(raw counts / nucleus total * 10000)) · referencePanel: Table S2 literature-derived Root column only · zScoreVariance: population · ambiguityMargin: 0.25 · statisticalTest: false · minimumEnrichment: 0.1 · minimumGeneZScore: 0.5 · minimumCandidateScore: 0.5 · minimumSupportingGenes: 2 · minimumAvailableFraction: 0.5 · minimumDetectionFraction: 0.1 ·

If retrieval or verification fails, annotation stops.

The candidate comes from measured gene activity.

Example: group 6, 388 nuclei

01 Normalise raw counts

log1p(count / cell total × 10,000)

02 Compare each group

Mean expression, detection and group-versus-rest

03 Score compatible markers

Across-group z-scores and published gene sets

04 Return a candidate

Root hair, supported by two measured genes



7 suggested groups / 1 ambiguous / 4 unresolved

The next answer remembers the reviewed decision.

Saved review

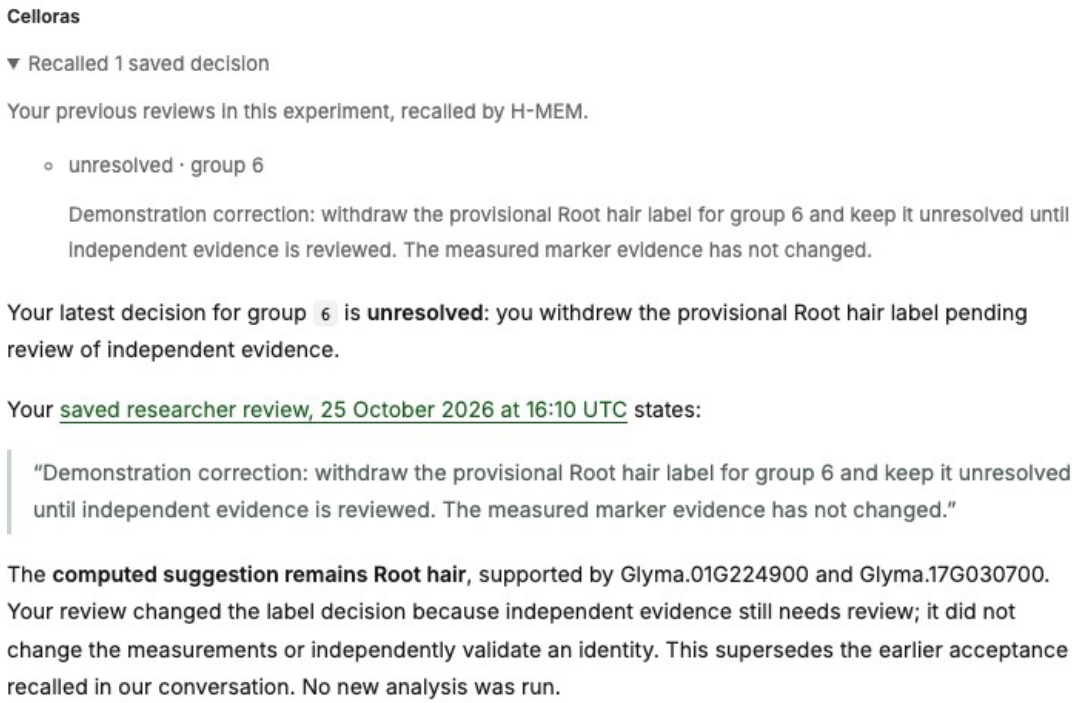
Root hair, provisional

Corrected review

Unresolved pending independent evidence

Next agent turn

Recall the latest review and cite its source



Celloras

▼ Recalled 1 saved decision

Your previous reviews in this experiment, recalled by H-MEM.

- unresolved · group 6

  Demonstration correction: withdraw the provisional Root hair label for group 6 and keep it unresolved until independent evidence is reviewed. The measured marker evidence has not changed.

Your latest decision for group `6` is **unresolved**: you withdrew the provisional Root hair label pending review of independent evidence.

Your saved researcher review, 25 October 2026 at 16:10 UTC states:

> "Demonstration correction: withdraw the provisional Root hair label for group 6 and keep it unresolved until independent evidence is reviewed. The measured marker evidence has not changed."

The **computed suggestion remains Root hair**, supported by Glyma.01G224900 and Glyma.17G030700. Your review changed the label decision because independent evidence still needs review; it did not change the measurements or independently validate an identity. This supersedes the earlier acceptance recalled in our conversation. No new analysis was run.

The correction changes the saved decision. The computed evidence stays intact.

Different data needs different storage.

Data	Where it lives	How the agent uses it
RNA counts and UMAP	Verified sparse files	Runs numerical tools on the matrix
Published marker genes	Zetaris reference table	Retrieves a checked reference version
Papers and guide	Curated passage files	Searches text and returns source links
Experiments and results	PostgreSQL	Reads only the authenticated project
Reviewed decisions	H-MEM + PostgreSQL	Recalls relevant decisions with receipts

Typed tool requests. Owned result IDs. Immutable saved results. No model-written SQL.

One workflow makes the judging criteria visible.

Criterion	What the video shows	Proof already recorded
Demo / 15 points	Cell map, marker evidence, review, recall	Complete public workflow, 25 Oct
Product & UX / 10 points	Clickable map and evidence cards in chat	Desktop and mobile browser checks
Sponsor tech / 10 points	Zetaris provenance + H-MEM correction recall	Real SQL receipt and sourced memory

NVIDIA OpenShell runs the CLI agent on the server.

The required pair is Zetaris + a Meterless Engine. Cursor and NVIDIA are optional.

H5AD upload is the next gap to close.

Works today

Published soybean root demo
Measured annotation and visual cards
Marker-list CSV imports
Zetaris reference and H-MEM reviews

Implementation in progress

Resumable H5AD uploads
Counts, group and genome validation
Private dataset attachment
A complete uploaded-file demo

Also open: independent biological validation and intermittent local CLI failures.

Demo source: verified GSE270392 counts converted to H5AD, preserving groups and UMAP.